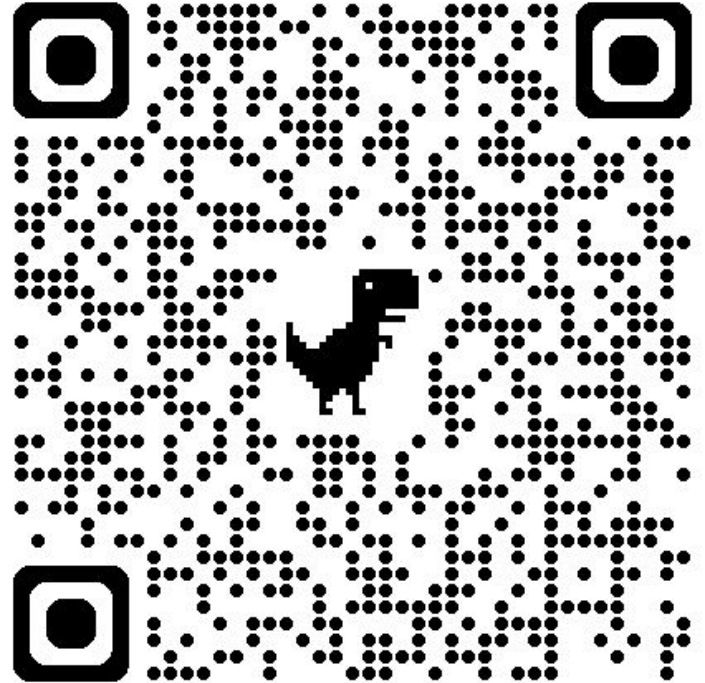


AI Agents Explained: Skills, Loops, and MCP

by Oleksandr Piekhota

Hard to read? Need links?



About me?

- Principal Software Dev at TeachingStrategies
- 14+ years in production engineering
- Backend-focused engineer (PHP, Go) and JavaScript enthusiast.
- MSc in CS & High-Performance Computing.
- Founder of TechTalks Edinburgh.



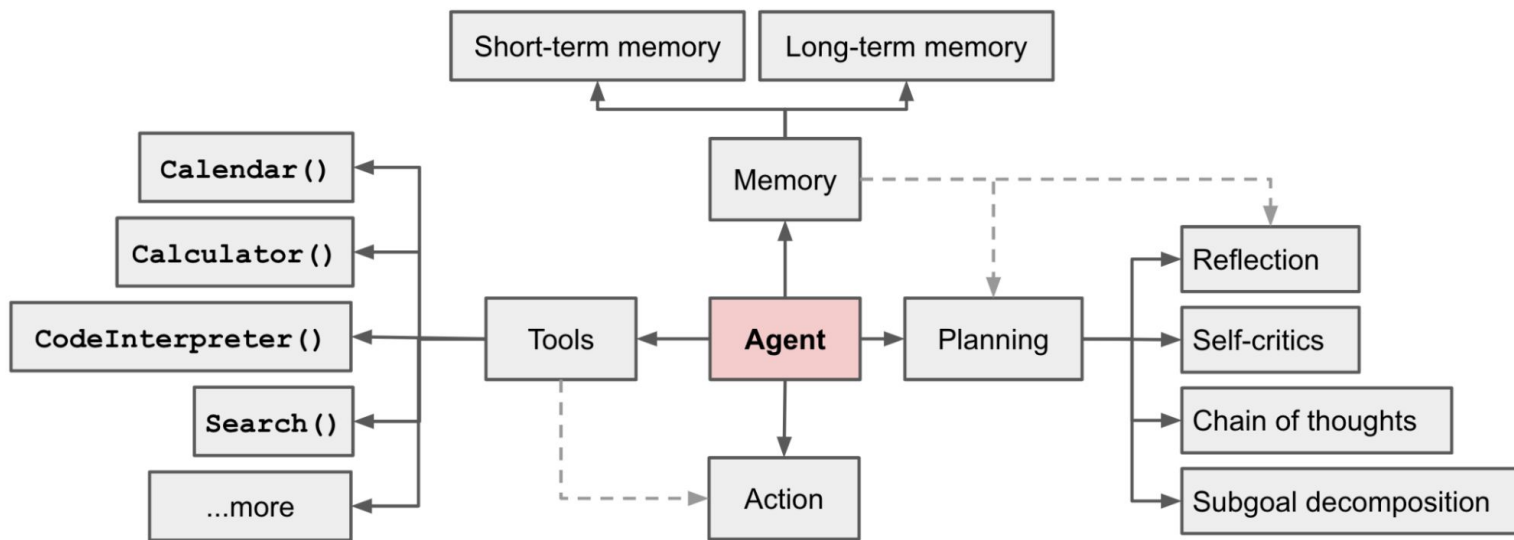
<https://www.linkedin.com/in/oleksandr-piekhota-b675ba53/>

Disclaimer

1. High Level Overview!
2. AI Agents - my personal experience and thoughts.
3. Difficult/impossible to follow/cover everything.
4. Use cases become more and more complicated - sub threads, agents that call other agents, mcp protocols nuances, token usage optimisations, etc

What is AI agent?

Agents are systems where LLMs dynamically direct their own processes and tool usage, maintaining control over how they accomplish tasks.



<https://www.anthropic.com/engineering/building-effective-agents>

<https://lilianweng.github.io/posts/2023-06-23-agent/>

Main AI Agent Components

1. Planning

- a. Subgoal and decomposition
- b. Reflection and refinement

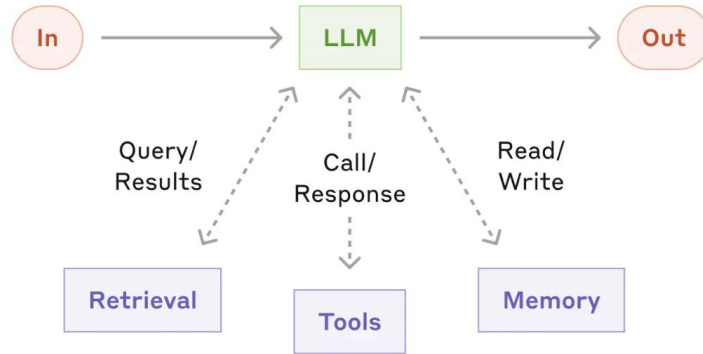
2. Memory

- a. Short-term memory
- b. Long-term memory (Vector search if large)

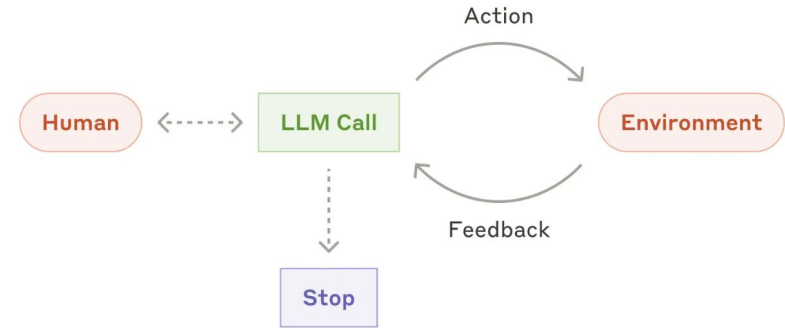
3. Tool Use - Action

- a. call external APIs for extra information
- b. RAG

How Agents Differ from Regular AI



The augmented LLM



Autonomous agent

Examples

Agentic tasks:

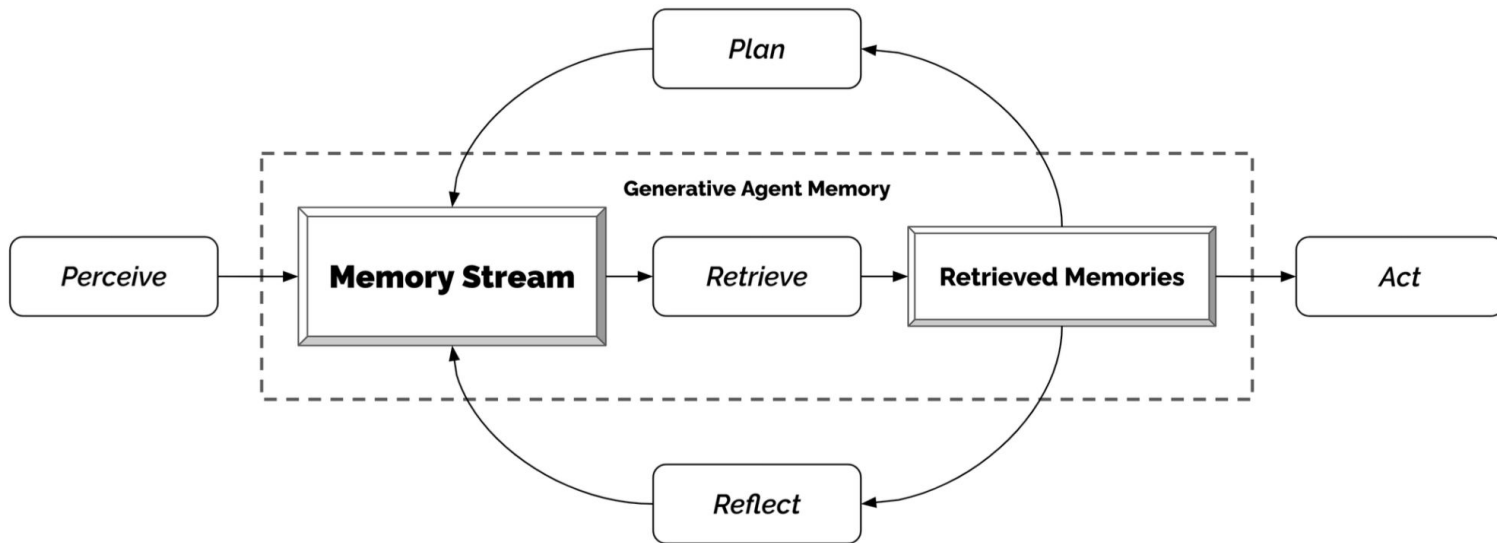
1. Summarize latest news about AI regulation
2. Check invoices in the current folder. Rename them according to the template {invoice_date}_{invoice_amount}_{company}.
3. Compare the current branch with development. Figure out changes and make code review.

Summarize latest news about AI regulation

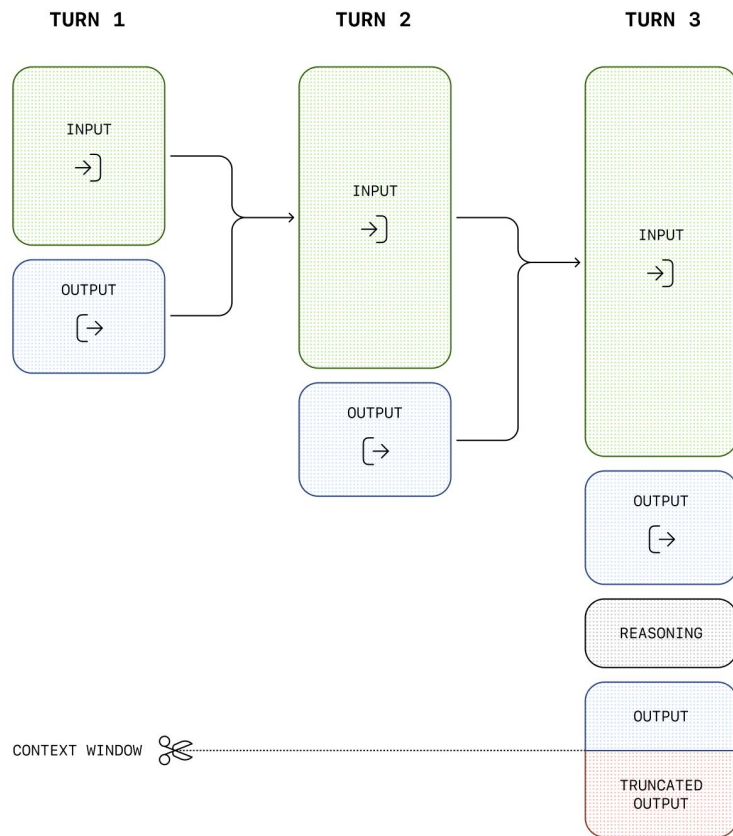
1. Calls a web search tool
2. Filters relevant articles
3. Extracts key points
4. Produces a structured summary

plan → fetch → process → respond

Repeated loop of actions



Context window



Context management compaction

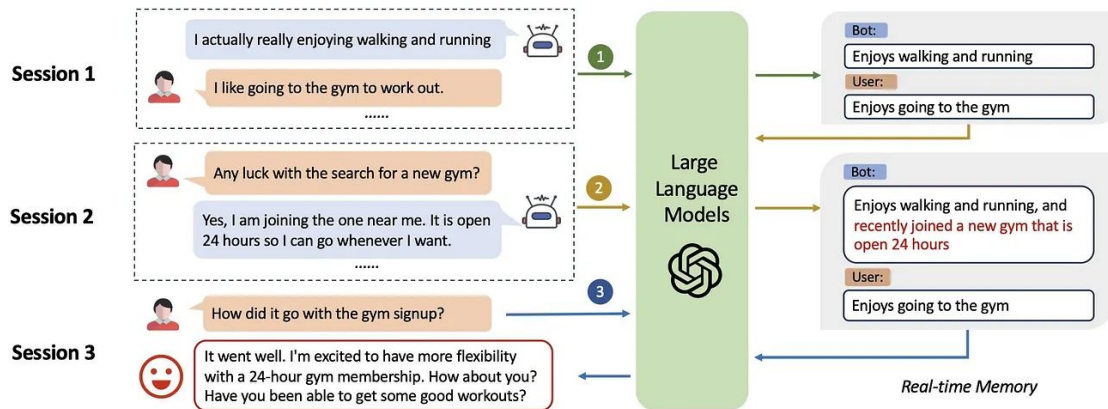


Figure 3: **The schematic overview of our method.** The model uses the first session to generate initial memory (green arrows), then updates the memory when the second session ends (yellow arrows), and generates a response using the latest memory at the third session (blue arrows).

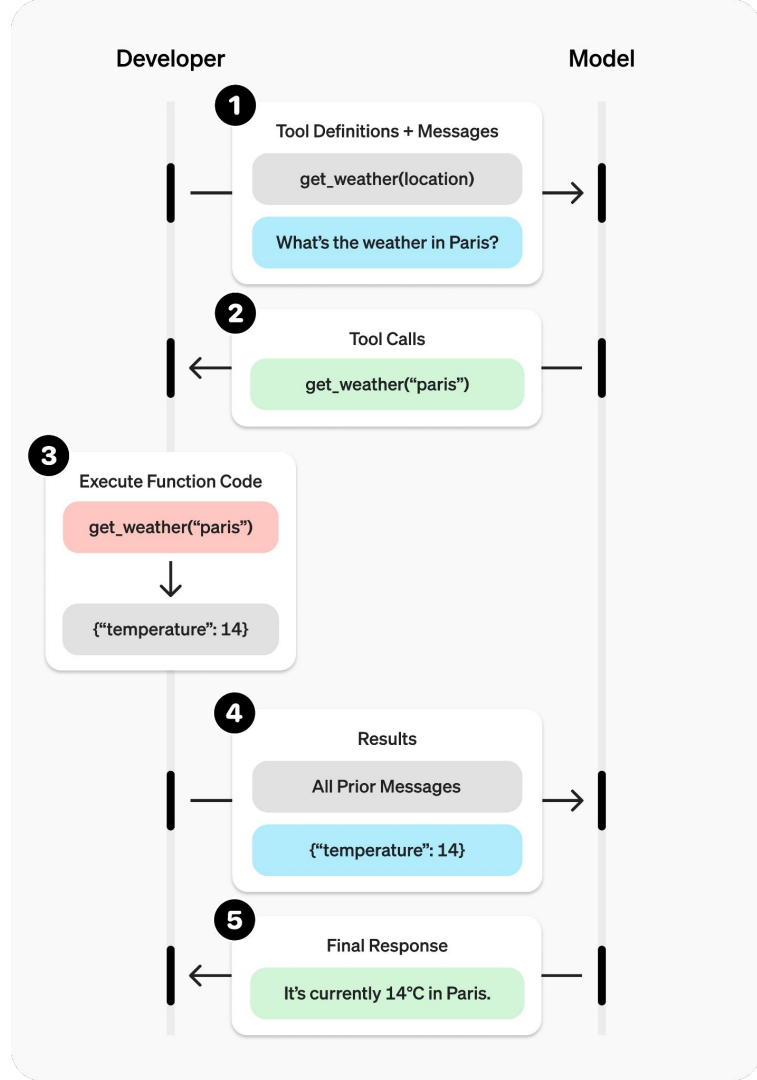
<https://kargarisaac.medium.com/the-fundamentals-of-context-management-and-compaction-in-llms-171ea31741a2>

<https://platform.claude.com/cookbook/tool-use-automatic-context-compaction>

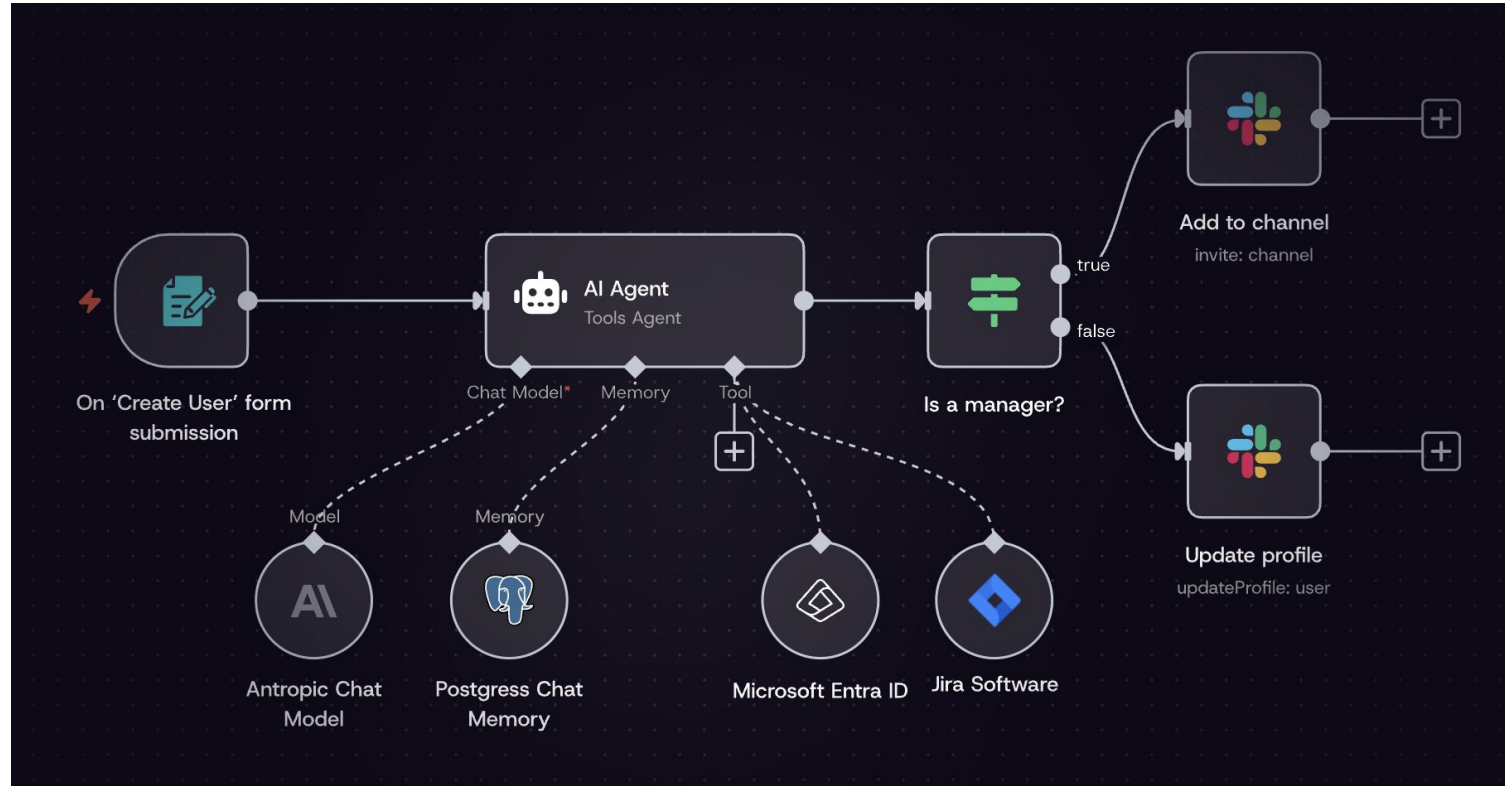
<https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents>

Tool calling (formerly function calling)

A function or tool refers in the abstract to a piece of functionality that we tell the model it has access to.



n8n and other automation software



Problem?

1. Hard to reuse
2. Hard to open source
3. Hard to contribute
4. Hard to adapt for multiple LLMs/Agents/etc.

Tool to access
FILE System

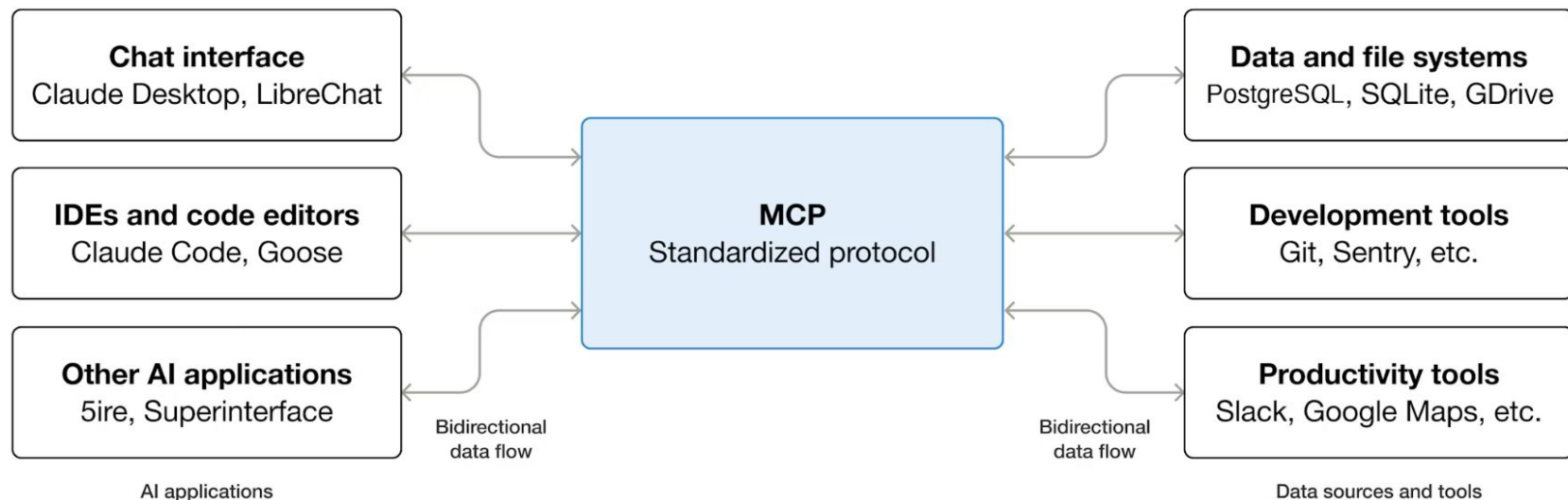
Tool to access
Browser

Tool to access
Database

Tool to access
OS

Tool to access
web service

Solution? - MCP by Anthropic (Nov 25, 2024)



MCP Example

1. Tools
2. Resources
3. Prompts

```
#[Name('User Server')]
#[Version('1.0.0')]
#[Instructions('This server exposes user management tools for the application.')]
class UserServer extends Server
{
    ...protected array $tools = [
        ...ListUsersTool::class,
        ...FindUserTool::class,
        ...CreateUserTool::class,
        ...UpdateUserTool::class,
        ...RenameUserTool::class,
        ...ChangeRoleTool::class,
    ];

    ...protected array $resources = [
        ...RolesResource::class,
        ...UserStatsResource::class,
        ...UsersResource::class,
    ];

    no usages

    ...protected array $prompts = [
        ...CreateUserPrompt::class,
        ...ManageUserPrompt::class,
        ...PromoteUserPrompt::class,
    ];
}
```

<https://github.com/pekhota/laravel-mcp-server-demo/pull/1>

MCP Inspector: Resources

MCP Inspector v0.21.2

Transport Type

Streamable HTTP

URL

http://localhost:8000/mcp/users

Connection Type

Via Proxy

Server Entry

Servers File

> Authentication

> Configuration

Reconnect

Disconnect

Connected

User Server

Version: 1.0.0

Resources

Prompts

Tools

Tasks

Apps

Ping

Sampling

Elicitations

Roots

Auth

Metadata

Resources

List Resources

Clear

roles-resource

user-stats-resource

users-resource

Resource Templates

List Templates

Clear

users-resource

Refresh

```
{
  contents: [
    0: {
      uri: "file://resources/users-r
esource"
      mimeType: "text/plain"
      text: "Total users: 6

- [1] Heloise Jacobs I <z
stanton@example.com> [vie
wer] (joined 2026-04-19)
- [2] Ransom Koepp <mitch
ell.leon@example.net> [vi
ewer] (joined 2026-04-19)
- [3] Frederick Okuneva D
DS <oziemmann@example.net>
[viewer] (joined 2026-04-
```

History

Clear

4. resources/read

3. resources/read

2. resources/list

1. initialize

Server Notifications

Clear

No notifications yet

MCP Inspector: Prompts

The screenshot displays the MCP Inspector v0.21.2 application interface. The left sidebar contains configuration options for Transport Type (Streamable HTTP), URL (http://localhost:8000/mcp/users), and Connection Type (Via Proxy). It also shows a list of servers, including 'User Server' (Version: 1.0.0), and a status bar indicating 'Connected'. The main panel is divided into two sections: 'Prompts' and 'History'. The 'Prompts' section lists three prompts: 'create-user-prompt', 'manage-user-prompt', and 'promote-user-prompt'. The 'create-user-prompt' is selected, showing its description and a form with fields for 'name' (John) and 'email' (Doe). The 'History' section shows a list of recent prompts, including 'prompts/get' and 'prompts/list'. The right sidebar displays the JSON representation of the selected prompt, including its description and messages.

MCP Inspector v0.21.2

Transport Type: Streamable HTTP

URL: http://localhost:8000/mcp/users

Connection Type: Via Proxy

Server Entry Servers File

> Authentication

> Configuration

Reconnect Disconnect

Connected

User Server
Version: 1.0.0

System

Prompts

List Prompts

Clear

create-user-prompt

Guides the creation of a new user by collecting their name, email, and password, then calling the create-user tool.

manage-user-prompt

Guides managing an existing user by looking them up and offering to update their name, email, or password.

promote-user-prompt

Guides promoting or demoting a user's role by looking them up, showing their current role, and applying the change.

History

Clear

7. prompts/get

6. prompts/get

5. prompts/list

create-user-prompt

Guides the creation of a new user by collecting their name, email, and password, then calling the create-user tool.

name

John

The full name of the user to create. (Required)

email

Doe

The email address of the user to create. (Required)

Get Prompt

```
{
  "description": "Guides the creation of a new user by collecting their name, email, and password, then calling the create-user tool.",
  "messages": [
    {
      "role": "assistant",
      "content": {
        "type": "text",
        "text": "You are a user management assistant. Your job is to create new users using the create-user tool. Always confirm the details before proceeding and report back with the created user's ID."
      }
    },
    {
      "role": "user",
      "content": {
        "type": "text",
        "text": "Please create a new user with the following details:
        - Name: John
        - Email: Doe

        Ask for a password if one has not been provided, then use the create-user tool to create the account."
      }
    }
  ]
}
```

<https://github.com/modelcontextprotocol/inspector>

MCP Inspector: Tools

MCP Inspector v0.21.2

Transport Type

Streamable HTTP

URL

http://localhost:8000/mcp/users

Connection Type

Via Proxy

Server Entry

Servers File

Authentication

Configuration

Reconnect

Disconnect

Connected

User Server

Version: 1.0.0

Resources

Prompts

Tools

Tasks

Apps

Ping

Sampling

Elicitations

Roots

Auth

Metadata

Tools

List Tools

Clear

List Users Tool

List users with optional search by name or email. Returns paginated results.

Find User Tool

Find a specific user by their ID or email address.

Create User Tool

Create a new user with a name, email, password, and optional role (defaults to viewer).

Update User Tool

Update a user's name, email, or password by their ID. At least one field must be provided.

Rename User Tool

Rename an existing user by their ID.

History

Clear

10. tools/call

9. tools/call

8. tools/list

7. prompts/get

Create User Tool

Create a new user with a name, email, password, and optional role (defaults to viewer).

Read-only

Destructive

Idempotent

Open-world

name *

Jogn

email *

john.doe@example.com

password *

123123123

role

admin

Tool-specific Metadata:

Add Pair

No metadata pairs.

Run Tool

Copy Input

Tool Result: Success

"User created successfully."

ID: 7

Name: Jogn

Email: john.doe@example.com

Role: admin"

.mcp.json

.mcp.json ×

```
1  {
2    "mcpServers": {
3      "laravel-boost": {
4        "command": "php",
5        "args": [
6          "artisan",
7          "boost:mcp"
8        ]
9      },
10     "user-server": {
11       "type": "http",
12       "url": "http://localhost:8000/mcp/users"
13     }
14   }
15 }
```

Claude

Example

- How can I help you?

Show me the list of current users

- user-server - list-users-tool (MCP)

- Here are the 7 current users:

ID	Name	Email	Role	Joined
7	Jogn	johndoe@example.com	admin	2026-04-22
6	Carol White	carol@example.com	viewer	2026-04-19
5	Bob Smith	bob@example.com	viewer	2026-04-19
4	Alice Johnson	alice@example.com	viewer	2026-04-19
1	HeLoise Jacobs I	zstanton@example.com	viewer	2026-04-19
2	Ransom Koepp	mittchell.leon@example.net	viewer	2026-04-19
3	Frederick Okuneva DDS	oziemann@example.net	viewer	2026-04-19

Note: User ID 7 "Jogn" appears to have a typo in the name – likely meant to be "John".

Change the role for mittchell.leon@example.net to admin

- user-server - change-role-tool (MCP)(id: 2, role: "admin")

- Done. Ransom Koepp (mittchell.leon@example.net) has been updated from viewer to admin.

show me update info for the user mittchell.leon@example.net

- user-server - find-user-tool (MCP)(email: "mittchell.leon@example.net")

- Here's the info for Ransom Koepp:

Field	Value
ID	2
Name	Ransom Koepp
Email	mittchell.leon@example.net
Role	admin
Verified	2026-04-19 20:02:38
Joined	2026-04-19 20:02:38
Updated	2026-04-22 14:30:23

The Updated timestamp (2026-04-22 14:30:23) reflects the recent role change to admin.

Claude Skills

Reusable instructions Claude can access when needed



ai-tutor/



saas-idea-validation/



frontend-design-audit/

Why?

Clear Instructions → Better Results

1) Write instructions every time

2) Search, copy, & paste from library



**Write instructions once.
Claude automatically
uses it when relevant**

Example

The image shows a code editor interface with a project file tree on the left and a code editor on the right. A red arrow points to the 'email-reminder' folder in the file tree, which contains a file named 'SKILL.md'. The code editor displays the content of 'SKILL.md' with a 'Meta' header and a 'Body' section.

Project File Tree:

- tech-talks-text ~/personal/tech-talks-text
 - .claude
 - skills
 - email-reminder
 - SKILL.md
 - linkedin-reminder
 - SKILL.md
 - meetup-post
 - SKILL.md
 - events
 - AGENTS.md
 - README.md
 - External Libraries
 - Scratches and Consoles

SKILL.md Content:

```
1 ---
2 name: email-reminder
3 description: This skill should be used when the user asks to "write an email reminder", "draft a meetup email", "create an event email", "generate TechT
4 version: 1.0.0
5 ---
6
7 # TechTalks Email Reminder
8
9 This skill generates reminder emails sent to registered attendees of upcoming TechTalks Edinburgh events via Meetup.com.
10
11 ## Structure
12
13 An email reminder has these sections in order:
14
15 1. **Subject line**
16 2. **Greeting**
17 3. **Venue change notice** (optional) – only when venue has changed
18 4. **Opening line** – date, time, excitement
19 5. **RSVP warning** (fixed)
20 6. **Location block**
21 7. **Venue change notice** (optional repeat)
22 8. **Parking** (optional)
23 9. **Food & Drinks**
24 10. **Photos & Videos**
25 11. **Venue change notice** (optional repeat)
26 12. **Sign-off**
27
```

How Skills Work

A folder with files (and folders) in it living in Claude's virtual machine

```
skill-name/  
├── SKILL.md  
├── how-to-x.md  
└── scripts  
    └── foo.py
```



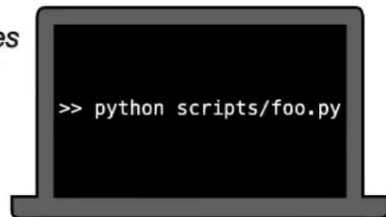
Code = Scripts that Claude can call via bash, Python, and Nodejs



Claude calls skill



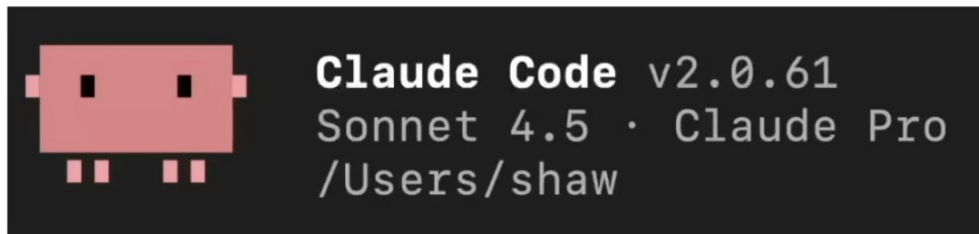
Executes script



(Virtual Machine)

Skills, MCP, & Subagents

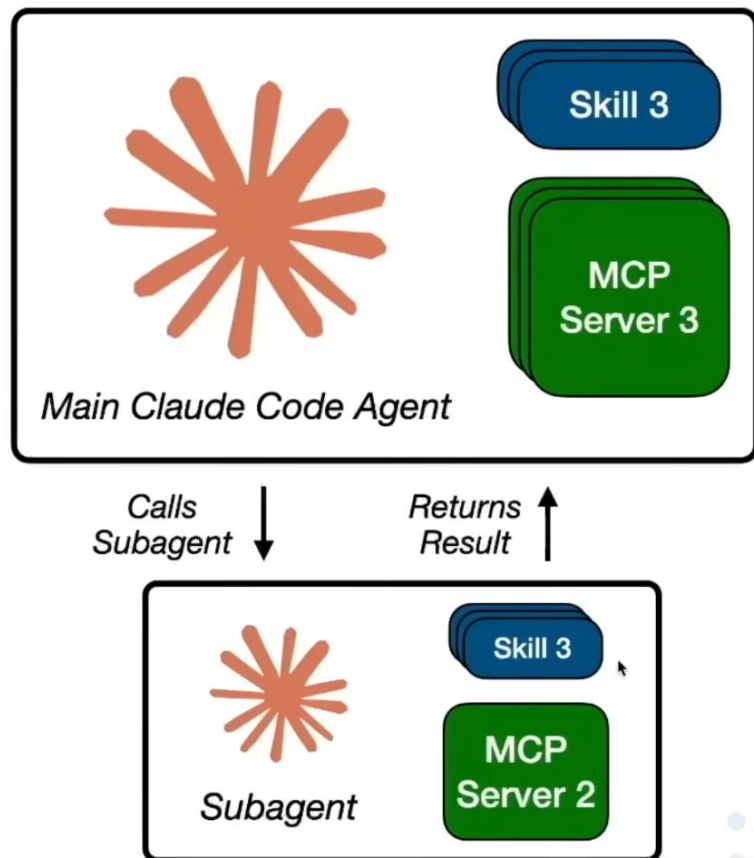
How these ideas fit together



Subagents = specialized **agents** for specific workflows (with their own context window)

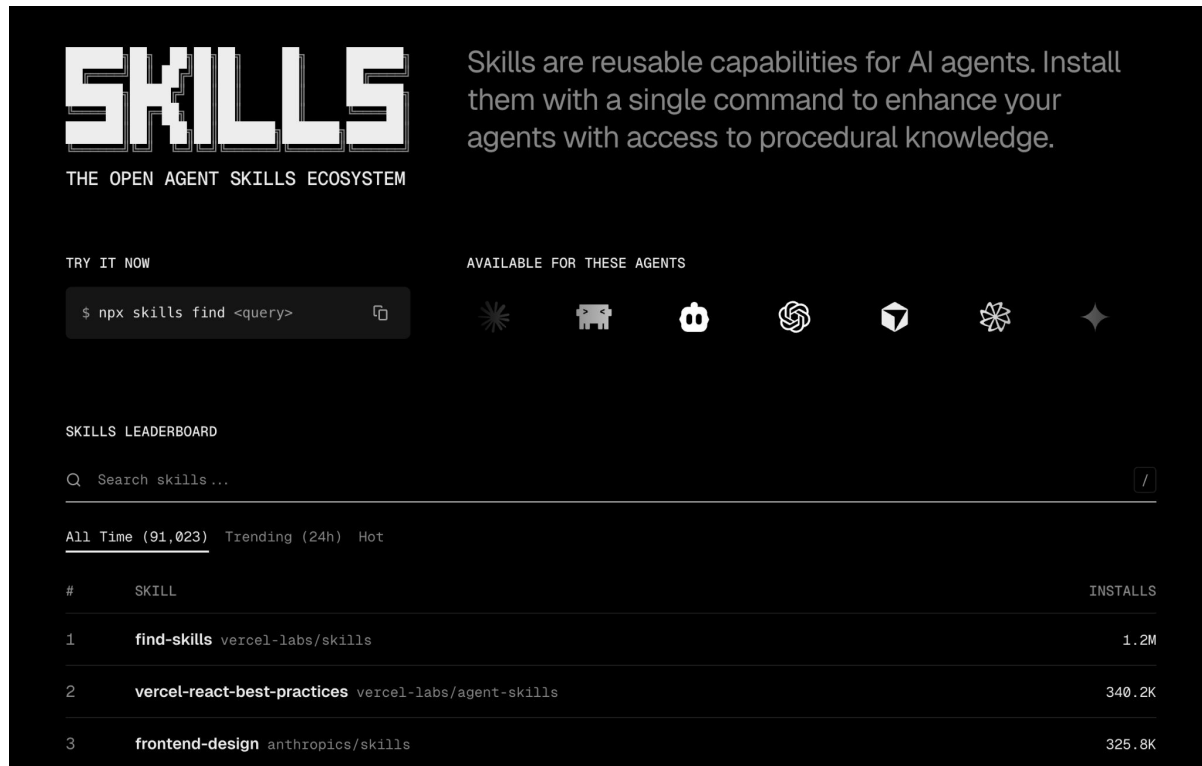
Skills = specialized **instructions/tools** for specific workflows

MCP Server = specialized **toolsets** for specific workflows



Skills are supported
by Claude, Codex,
and other agents

<https://skills.sh/> - skills marketplace



The screenshot shows the Skills marketplace website. At the top, the 'SKILLS' logo is displayed in a large, stylized font, with the tagline 'THE OPEN AGENT SKILLS ECOSYSTEM' below it. To the right, a text block states: 'Skills are reusable capabilities for AI agents. Install them with a single command to enhance your agents with access to procedural knowledge.'

Below the header, there are two main sections. The first is 'TRY IT NOW', which features a dark button with the command `$ npx skills find <query>` and a copy icon. The second is 'AVAILABLE FOR THESE AGENTS', which displays a row of logos for various AI agents: a starburst, a robot head, a dog head, a hexagon, a flower, and a diamond.

The bottom section is the 'SKILLS LEADERBOARD'. It includes a search bar with the placeholder text 'Search skills...'. Below the search bar, there are three tabs: 'All Time (91,023)', 'Trending (24h)', and 'Hot'. The 'All Time' tab is selected. The leaderboard table has three columns: '#', 'SKILL', and 'INSTALLS'. The top three skills are listed as follows:

#	SKILL	INSTALLS
1	find-skills vercel-labs/skills	1.2M
2	vercel-react-best-practices vercel-labs/agent-skills	340.2K
3	frontend-design anthropics/skills	325.8K

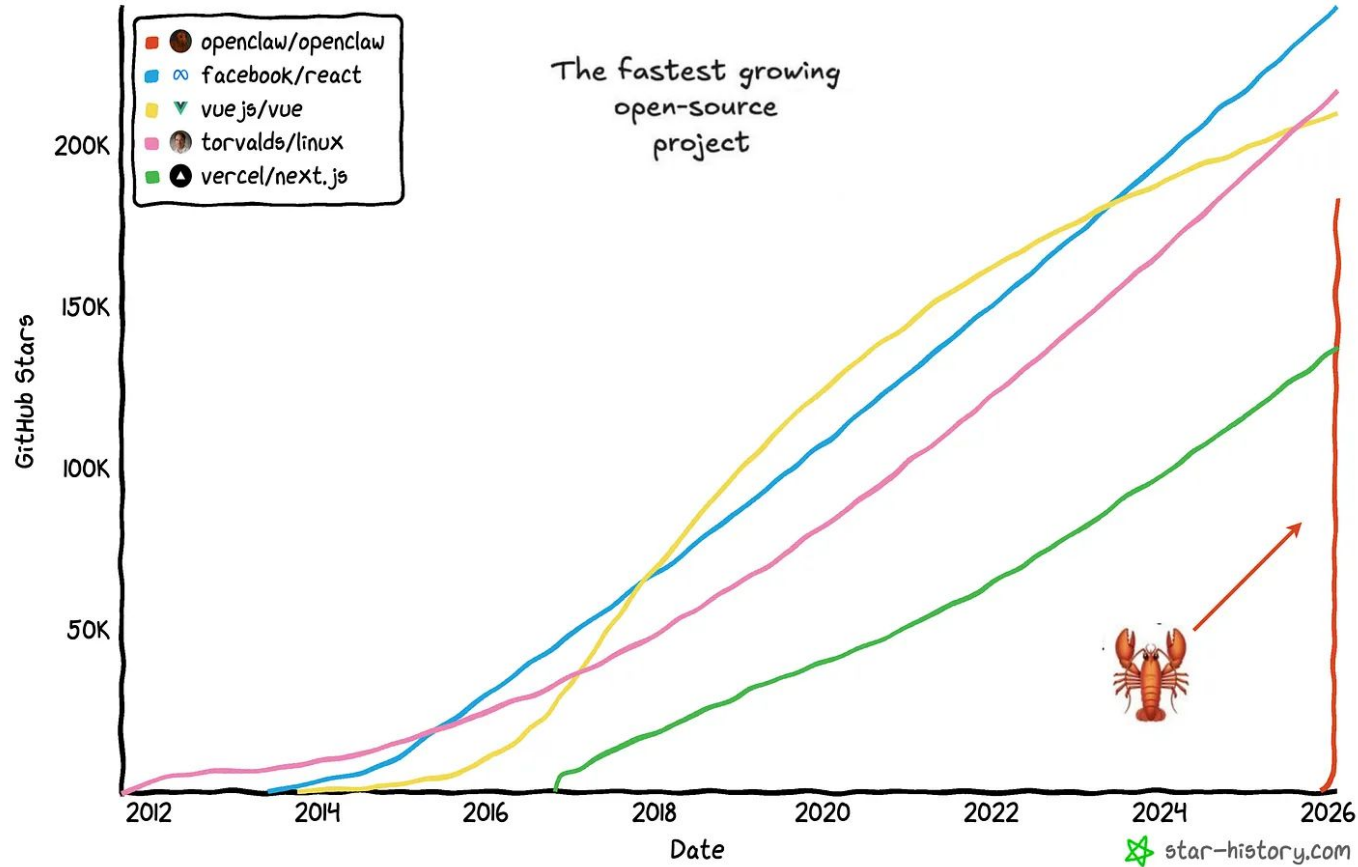
<https://github.com/garrytan/gstack/tree/main> - Use Garry Tan's exact Claude Code setup: 23 opinionated tools that serve as CEO, Designer, Eng Manager, Release Manager, Doc Engineer, and QA



OpenClaw

THE AI THAT ACTUALLY DOES THINGS.

Star History





PicoClaw: Ultra-Efficient AI Assistant in Go

\$10 Hardware · 10MB RAM · ms Boot · Let's Go, PicoClaw!

[Go](#) 1.25+
 [Arch](#) x86_64, ARM64, MIPS, RISC-V, LoongArch
 [license](#) MIT

[Website](#) [pico claw.io](#)
[Docs](#) [Official](#)
[Wiki](#) [DeepWiki](#)

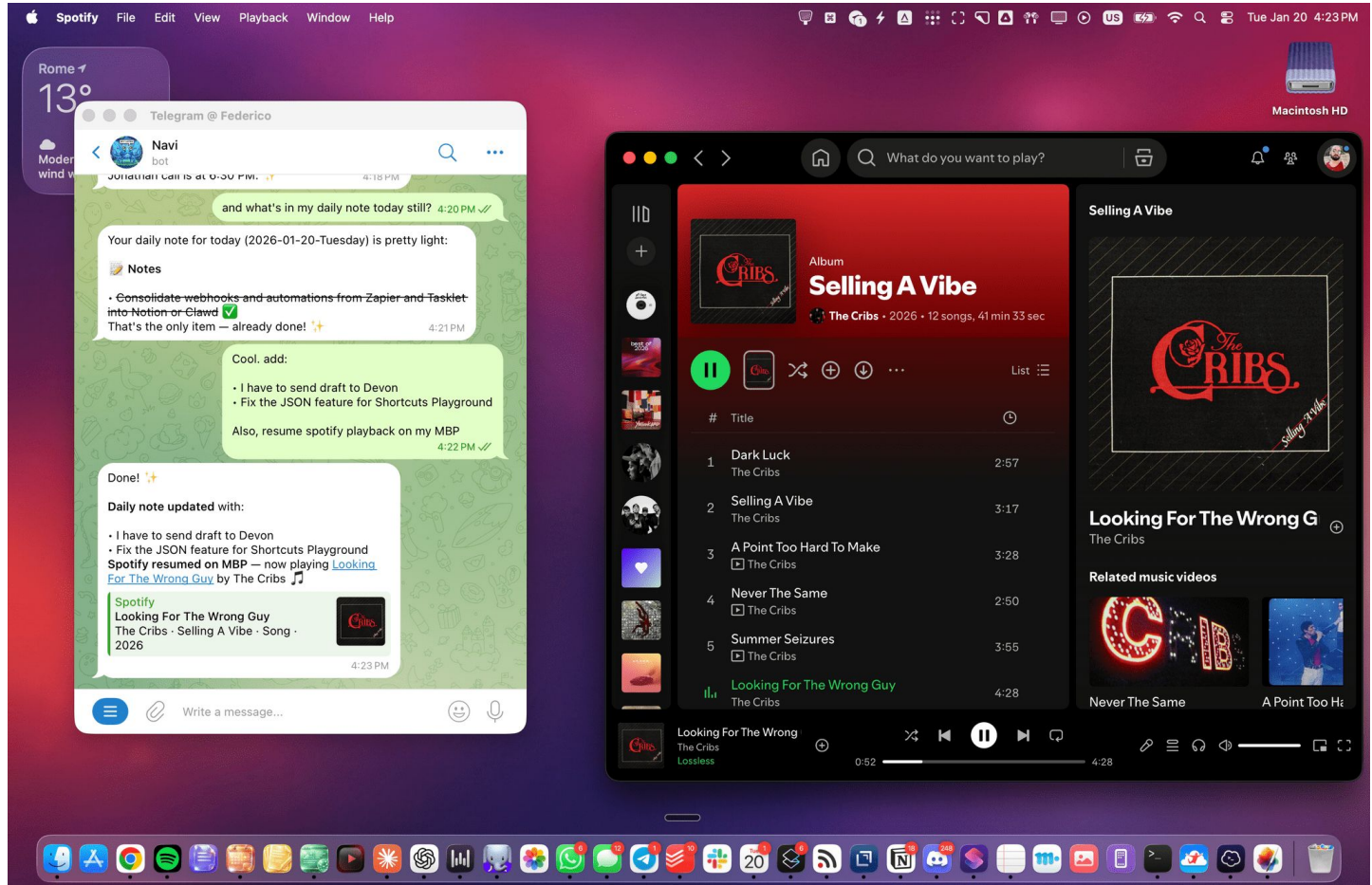
[X \(Twitter\)](#) [SpeedIO](#)
[WeChat](#) [Group](#)
[Discord](#) [Community](#)

[中文](#) | [日本語](#) | [한국어](#) | [Português](#) | [Tiếng Việt](#) | [Français](#) | [Italiano](#) | [Bahasa Indonesia](#) | [Malay](#) | [English](#)

Raspberry Pi Holdings

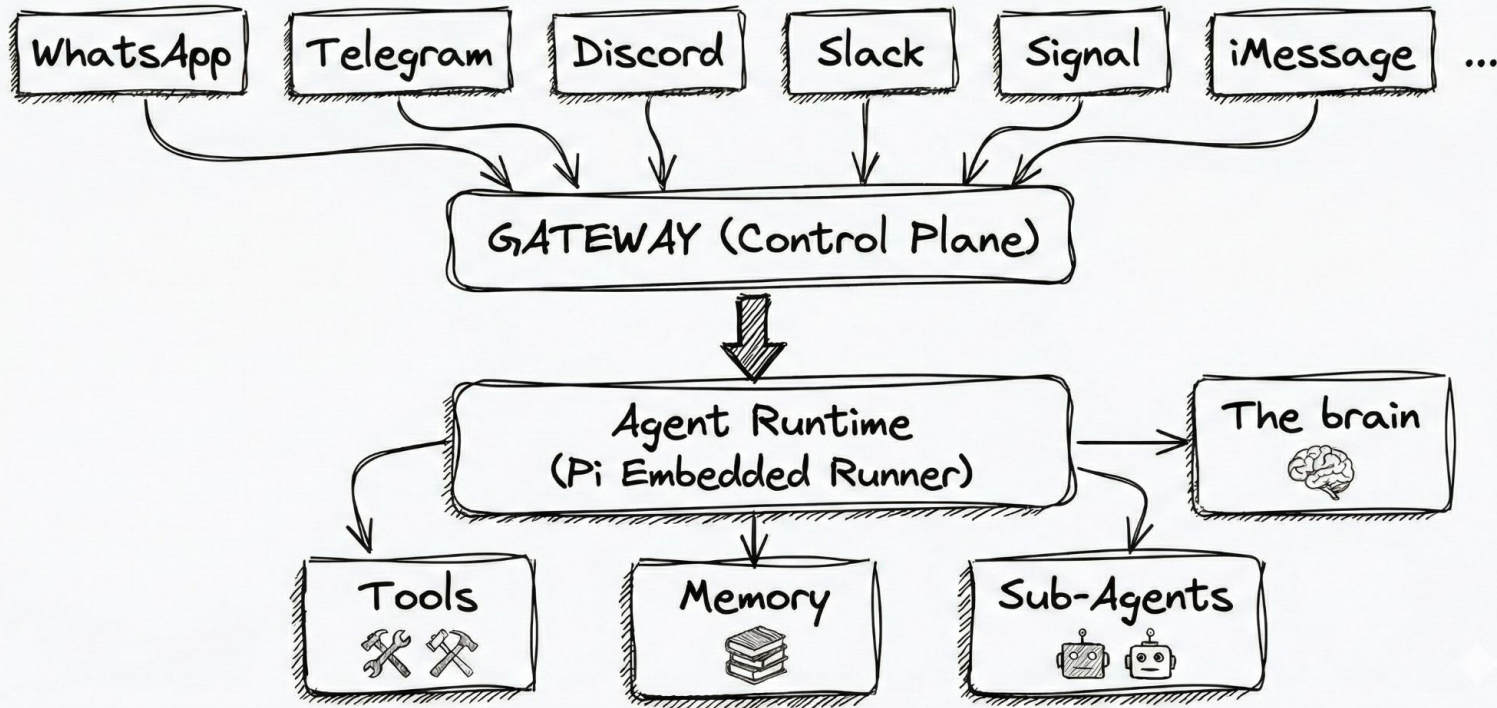


What is OpenClaw



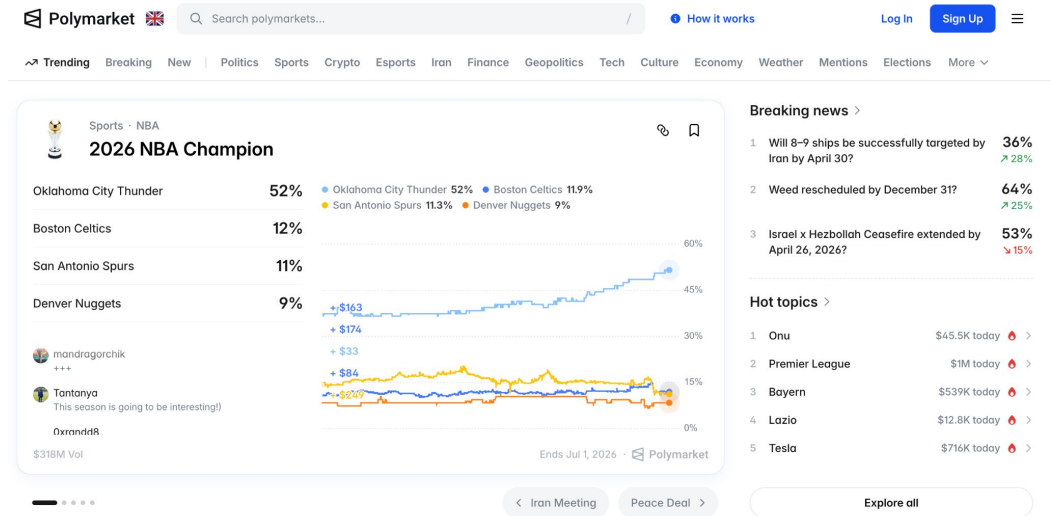
<https://www.macstories.net/stories/clawdbot-showed-me-what-the-future-of-personal-ai-assistants-looks-like/>

OpenClaw Architecture (MoltBot)



CRON & Scheduled tasks + Possible use cases

- Social Intent Research & Polymarket Bets
- Crypto Research & Crypto Whales following trading
- Reminders and automations
- etc



<https://docs.openclaw.ai/automation>

<https://www.macstories.net/stories/clawdbot-showed-me-what-the-future-of-personal-ai-assistants-looks-like/>

@mariozechner/pi-agent-core 

0.67.67 • Public • Published an hour ago

 Readme

 Code 

 1 Dependency

 395 Dependents

 218 Versions

@mariozechner/pi-agent-core

Stateful agent with tool execution and event streaming. Built on @mariozechner/pi-ai.

Installation

```
npm install @mariozechner/pi-agent-core
```

Quick Start

```
import { Agent } from "@mariozechner/pi-agent-core";
import { getModel } from "@mariozechner/pi-ai";

const agent = new Agent({
  initialState: {
    systemPrompt: "You are a helpful assistant.",
    model: getModel("anthropic", "claude-sonnet-4-20250514"),
  },
});

agent.subscribe((event) => {
  if (event.type === "message_update" && event.assistantMessageEvent.type === "text") {
    // Stream just the new text chunk
    process.stdout.write(event.assistantMessageEvent.delta);
  }
});

await agent.prompt("Hello!");
```

Install

```
> npm i @mariozechner/pi-agent-core
```

Repository

 github.com/badlogic/pi-mono

Homepage

 github.com/badlogic/pi-mono#readme

Weekly Downloads

1,962,971

Version

0.67.67

License

MIT

Last publish

an hour ago

Collaborators



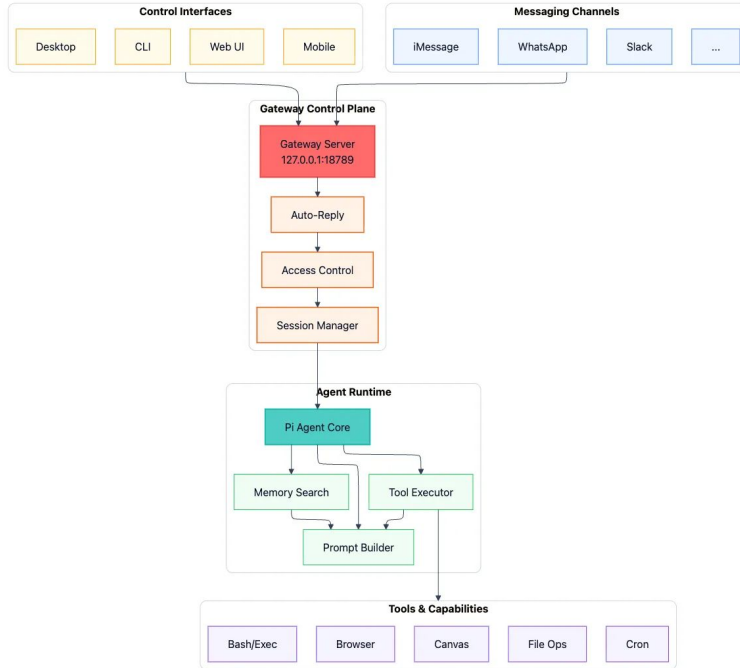
 Analyze security with Socket

 Check bundle size

There are plenty of agents available. Not only claude or codex...

- <https://openai.com/codex/>
- <https://code.claude.com/docs/en/overview>
- <https://docs.langchain.com/oss/python/langchain/agents>
- <https://pi.dev/>
- <https://github.com/microsoft/semantic-kernel>
- <https://github.com/huggingface/smolagents>
- <https://opencode.ai/>

OpenClaw Extra to Read and Watch (Peter Steinberger)



<https://www.youtube.com/watch?v=YFjfBk8HI5o>

<https://ppaolo.substack.com/p/openclaw-system-architecture-overview>

What's important?

- MCP extension power - ClawHub

The screenshot displays the ClawHub website interface. At the top, there's a navigation bar with the ClawHub logo, a search bar, and a 'Sign in with GitHub' button. Below the navigation bar, there are tabs for 'Skills', 'Plugins', 'Users', and 'About'. The main content area features three categories: 'Skills' (Agent skill bundles), 'Plugins' (Gateway plugins), and 'Builders' (Community creators). A statistics bar shows '52.7k tools', '180k users', '12M downloads', and '4.8 avg rating'. The 'TRENDING NOW' section lists six items, each with a description, author, and an 'Install' button.

Item	Author	Description	Stars	Downloads	Install Button
self-improving-agent	by pskoett	Captures learnings, errors, and corrections to enable continuous improvement. Use when: (1) A command or operation fails...	3.3k	406.4k	Install
Self-Improving + Proactive Agent	by ivangdavila	Self-reflection + Self-criticism + Self-learning + Self-organizing memory. Agent evaluates its own work, catches mistakes, and improves...	990	170.4k	Install
ontology	by oswalpalash	Typed knowledge graph for structured agent memory and composable skills. Use when creating/querying entities (Person, Project, Task,...	556	169.5k	Install
Polymarket	by joelchance	Query Polymarket prediction markets. Check odds, find trending markets, search events, track price movements.	88	130.0k	Install
Multi Search Engine	by gpyangyoujun	Multi search engine integration with 16 engines (7 CN + 9 Global). Supports advanced search operators, time filters, site search, privacy...	591	125.9k	Install
AdMapix	by flyOpants	Ad intelligence & app analytics assistant. Search ad creatives, analyze apps, view rankings, track downloads/revenue, and get market insight...	240	96.4k	Install

Tokens are getting more expensive!

NEWS

[UPDATED] News: Anthropic (Briefly) Removes Claude Code From \$20-A-Month "Pro" Subscription Plan For New Users

Model	Tokens	vs. lowest
claude-opus-4-7	7,335	1.46x
claude-opus-4-6	5,039	1.00x

Opus 4.7 uses the same pricing as Opus 4.6 - \$5 per million input tokens and \$25 per million output tokens - but this token inflation means we can expect it to be around 40% more expensive.

<https://www.wheresyoured.at/news-anthropic-removes-pro-cc/>

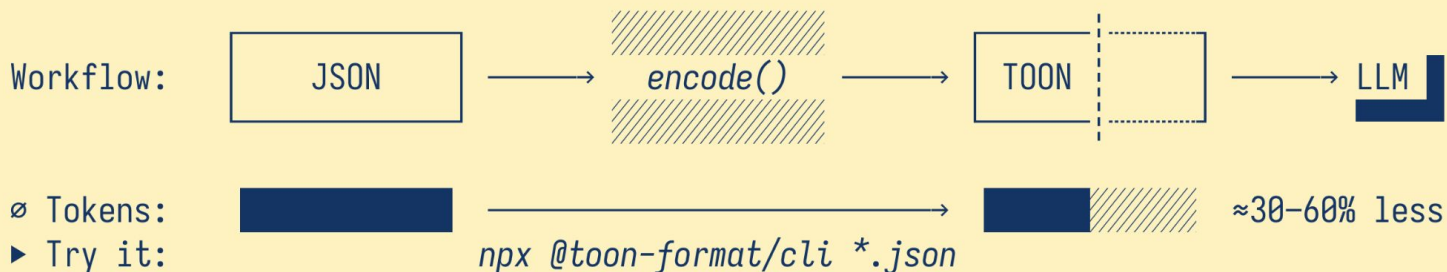
<https://simonwillison.net/2026/apr/20/claude-token-counts/>

Token optimisation techniques and libraries - Token-Oriented Object Notation (TOON)

Token-Oriented Object Notation



Compact, human-readable serialization of JSON data for LLM prompts



	avg tokens	retrieval accuracy	best for
TOON →		73.9%	repeated structure • tables
JSON →		69.7%	varying fields • deep trees

Questions?

